

Vector Analysis

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Color Codes

- ▶ *Black*: regular text.
- ▶ *Red*: important concept, emphasis,...
- ▶ *Green*: *optional*, but worth thinking about.
- ▶ *Blue*: required HW, test your understanding!
- ▶ *Orange*: in jest; take with a pinch of salt!

Applications of Electromagnetism



- Define electromagnetism.

What is Electromagnetism?

- ▶ *Electromagnetism* is a branch of physics which describes the **interaction** between **charged particles**.
- ▶ Charges come in only **two** flavors: positive (+), and negative (−).
- ▶ By **interaction** we mean the (electromagnetic) **forces** which the charges exert on each other.
- ▶ The EM force is '*carried/mediated*' by **electromagnetic fields** composed of the electric fields (***E***) and magnetic fields (***B***).

Fields

- ▶ Fields?
- ▶ **Simplified:** A **field** is a physical quantity, represented by a number, or a vector that has a **value** for each point in **space-time**, *i.e.*, (x, y, z, t) .
 1. *e.g.*, pressure (*scalar field*)
 2. *e.g.*, wind velocity (as seen on a weather report) (*vector field*)
 3. *e.g.*, **the E and B fields of electromagnetism.** (*vector fields*)
- ▶ **More precisely:** A **field** is a physical quantity, represented by a **tensor**¹ (*e.g.*, a number is a rank-0 tensor, a vector is a rank-1 tensor), that has a **value** for each point in **space-time**, *i.e.*, (x, y, z, t) .
 - ▶ *e.g.*, **stress tensor** (*rank-2 tensor field*)

¹*def:* a **tensor** is an algebraic object (*e.g.*, vector or scalar or other tensors) that describes a **linear mapping** from one set of algebraic objects to another.

How are Electric and Magnetic Fields Produced?

- ▶ How are the electric and magnetic fields produced?
- ▶ A **stationary** charged particle: produces a *static electric field*.
- ▶ Similarly, a **steady current** in a wire (**also**) produces a *static magnetic field*.
- ▶ An **accelerating** charged particle produces an **electromagnetic (EM) wave/radiation**, *i.e.*, a synchronized oscillation of electric and magnetic fields which have the property that they travel through empty space/vacuum at the **speed of light** c . (*e.g.*, light, X-rays)

The Theoretical Basis of Classical Electromagnetism

- Only these 4 Maxwell's equations in the SI unit convention,

$$\nabla \cdot \mathbf{D} = \rho_f, \quad (\text{Gauss' law of electricity})$$

$$\nabla \cdot \mathbf{B} = 0, \quad (\text{Gauss' law of magnetism})$$

$$-\nabla \times \mathbf{E} = \frac{\partial \mathbf{B}}{\partial t}, \quad (\text{Faraday's law of induction})$$

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}, \quad (\text{Ampere's law})$$

where ρ_f is the free charge density and, $\mathbf{J}_f$ is the free current density.
... combined with the **Lorentz force law**

$$\mathbf{F} = q\mathbf{E} + q(\mathbf{v} \times \mathbf{B})$$

, form the basis of **classical** electromagnetism, and optics.

The Theoretical Basis of Classical Electromagnetism

- ▶ **Maxwell's equations** in the *SI unit convention*, constitute a set of 4 equations for both $[E (\mathbf{D})]$ and $[B (\mathbf{H})]$ which are...
 - ▶ *coupled*,
 - ▶ *first-order*,
 - ▶ *partial differential*

Vectors vs. Scalars

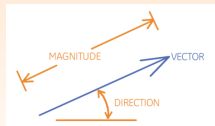


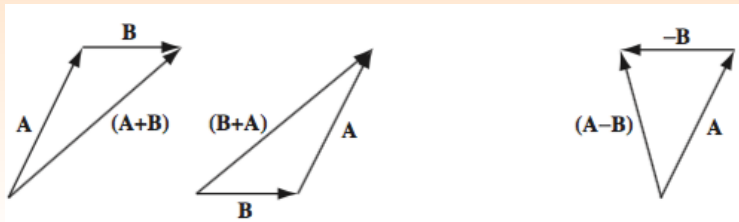
Figure: [Image from kullabs.com]

- ▶ **Vector** (*noun*): a quantity having a **direction** as well as a **magnitude**.
- ▶ *e.g.*, velocity, acceleration, force and momentum.
- ▶ **Scalar** (*noun*): a quantity that has a magnitude, but no direction.
- ▶ *e.g.*, mass, charge, density, and temperature.
- ▶ **Abstract mathematical** vectors have magnitude, and direction, but not *location*.
- ▶ *Typographically*, vectors shall be represented by a **bold face**, *e.g.*, **A**.

Vector Operations

- ▶ We will encounter **four** kinds of **vector operations**:
 - ▶ one *addition* ($\mathbf{A} + \mathbf{B}$),
 - ▶ and three kinds of '*multiplication*'.
 - ▶ Multiplication by a scalar ($k\mathbf{A}$),
 - ▶ Dot product of two vectors ($\mathbf{A} \cdot \mathbf{B}$),
 - ▶ Cross product of two vectors ($\mathbf{A} \times \mathbf{B}$).

Vector Addition



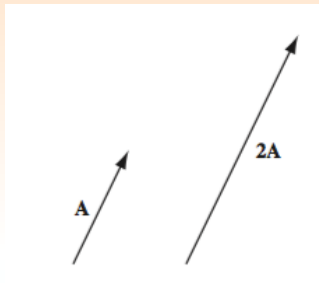
- ▶ For *graphically* representing vector addition, recall the *triangle rule* (from high school).
- ▶ Vector addition is **commutative**, *i.e.*,

$$A + B = B + A$$

- ▶ Vector addition is also **associative**, *i.e.*,

$$(A + B) + C = A + (B + C)$$

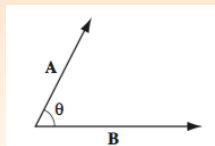
Multiplication by a scalar



- ▶ Multiplication of a vector by a positive scalar a multiplies the magnitude but leaves the direction unchanged.
- ▶ Scalar multiplication is **distributive**, *i.e.*,

$$a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$$

Dot Product of Two Vectors



- ▶ The **dot product** of two vectors is *defined* ($\equiv$) by,

$$\mathbf{A} \cdot \mathbf{B} \equiv |\mathbf{A}| |\mathbf{B}| \cos \theta$$

where θ is the angle they form *when* placed tail-to-tail.

- ▶ $\mathbf{A} \cdot \mathbf{B}$ yields a **scalar**, hence the *alternative* name **scalar product**.
- ▶ The dot product is **commutative**, *i.e.*,

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$$

- ▶ The dot product is **distributive**, *i.e.*,

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$$

Dot product of two vectors

- ▶ Given vectors in **component form**, *i.e.*,

$$\mathbf{A} = a_1 \hat{x}_1 + a_2 \hat{x}_2 + a_3 \hat{x}_3 + \dots = \sum_i a_i \hat{x}_i$$

$$\mathbf{B} = b_1 \hat{x}_1 + b_2 \hat{x}_2 + b_3 \hat{x}_3 + \dots = \sum_i b_i \hat{x}_i$$

- ▶ Component-wise we can *define* the dot product as,

$$\mathbf{A} \cdot \mathbf{B} \equiv a_1 b_1 + a_2 b_2 + a_3 b_3 + a_4 b_4 + \dots \equiv \sum_i a_i b_i$$

- ▶ Also note that,

$$\mathbf{A} \cdot \mathbf{A} = |\mathbf{A}|^2 = A^2$$

where A^2 or $|\mathbf{A}|^2$ are just **shorthands** for $\mathbf{A} \cdot \mathbf{A}$.

- ▶ and,

$$\mathbf{0} \cdot \mathbf{A} = 0$$

Orthogonality and Projections

- ▶ Two vectors $\mathbf{A}$ and $\mathbf{B}$ are **orthogonal**² if and only if,

$$\mathbf{A} \cdot \mathbf{B} = 0$$

- ▶ The **scalar projection**³ of $\mathbf{B}$ onto $\mathbf{A}$ is defined as,

$$P_{\mathbf{B},\mathbf{A}} = \frac{\mathbf{B} \cdot \mathbf{A}}{A}$$

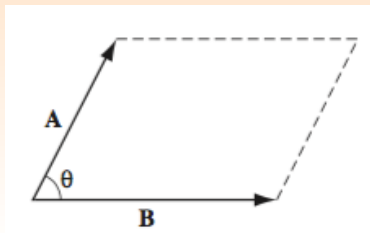
- ▶ The **vector projection** of $\mathbf{B}$ onto $\mathbf{A}$ is defined as,

$$\mathbf{P}_{\mathbf{B},\mathbf{A}} = \frac{\mathbf{B} \cdot \mathbf{A}}{A^2} \mathbf{A}$$

²The term **perpendicular** describes a property of two vectors, **orthogonal** is a related property of any collection of vectors (*i.e.*, a collection of vectors is orthogonal if and only if all of them are *pairwise* perpendicular), and **normal** is a relation between a vector and an object such as the tangent plane at a point of a smooth surface.

³Is $P_{\mathbf{A},\mathbf{B}} = P_{\mathbf{B},\mathbf{A}}$?

Cross Product of Two Vectors



- ▶ The **cross product** of two vectors is *defined* by,

$$\mathbf{A} \times \mathbf{B} \equiv |\mathbf{A}| |\mathbf{B}| \sin \theta \, \hat{\mathbf{n}}$$

where θ is the angle they form when placed tail-to-tail, and $\hat{\mathbf{n}}$ is a unit vector⁴ pointing $\perp$ to the plane of $\mathbf{A}$ and $\mathbf{B}$.

- ▶ The correct *orientation* of $\hat{\mathbf{n}}$ is determined by the **right-hand rule**, e.g., $\mathbf{A} \times \mathbf{B}$ above points into the page.

⁴a hatted $\hat{\mathbf{n}}$ denotes a unit vector

Cross Product of Two Vectors

- ▶ $\mathbf{A} \times \mathbf{B}$ is itself a vector (hence the alternative name **vector product**).
- ▶ The cross product is **distributive**,

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C})$$

- ▶ But the cross product is **not commutative**,

$$\mathbf{B} \times \mathbf{A} \neq \mathbf{A} \times \mathbf{B}$$

and instead,

$$\mathbf{B} \times \mathbf{A} = -(\mathbf{A} \times \mathbf{B})$$

- ▶ *Geometrically*, $|\mathbf{A} \times \mathbf{B}|$ is the **area** of the parallelogram *generated* by $\mathbf{A}$ and $\mathbf{B}$.

Cross Product of Two Vectors

- ▶ Given component-wise **3d** vectors we can conveniently *calculate* the cross product via the **determinant**⁵.

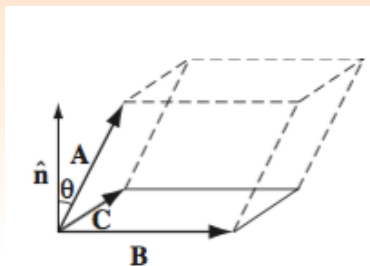
$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x}_1 & \hat{x}_2 & \hat{x}_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}$$

- ▶ Expand the above determinant and check that it matches with the expression 13, Chapter I of [IEDJ].
- ▶ The vector $\mathbf{A} \times \mathbf{B}$ is *orthogonal* to both $\mathbf{A}$ and $\mathbf{B}$.
- ▶ Two non-zero vectors $\mathbf{A}$ and $\mathbf{B}$ are parallel/anti-parallel *if and only if*

$$\mathbf{A} \times \mathbf{B} = 0$$

⁵You must be able to calculate determinants for both 3×3 , and 2×2 matrices.

Scalar Triple Product



- ▶ The **scalar triple product** between vectors A , B and C is given by

$$A \cdot (B \times C)$$

- ▶ Geometrically, $A \cdot (B \times C)$ is the **volume of the parellopiped** generated by A , B and C .
- ▶ Cyclic order preserves sign, i.e.,

$$A \cdot (B \times C) = B \cdot (C \times A) = C \cdot (A \times B)$$

Scalar Triple Product

- ▶ While non-cyclic permutations **reverse sign**, *i.e.*,

$$\mathbf{A} \cdot (\mathbf{C} \times \mathbf{B}) = \mathbf{B} \cdot (\mathbf{A} \times \mathbf{C}) = \mathbf{C} \cdot (\mathbf{B} \times \mathbf{A}) = -\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$$

- ▶ In component form, the scalar triple product evaluates to a **determinant**,

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$

- ▶ The dot and cross can be *interchanged* (keeping the same cyclic order),

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$$

- ▶ **Problem:** What's wrong with $(\mathbf{A} \cdot \mathbf{B}) \times \mathbf{C}$?

Vector Triple Product

- ▶ The **vector triple product** between vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ is given by

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$$

which geometrically amounts to?

- ▶ ... "a vector in the plane spanned by $\mathbf{B}$ and $\mathbf{C}$, which is also perpendicular to $\mathbf{A}$ ".
- ▶ It can be simplified via the **BAC-CAB identity**,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \equiv \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

- ▶ Let's *regroup* the **brackets**. Since cross-products are **not associative**, i.e.,

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \neq \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$$

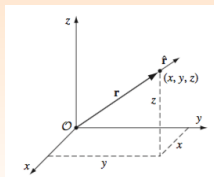
it turns out that this *regrouping*,

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = -\mathbf{A}(\mathbf{B} \cdot \mathbf{C}) + \mathbf{B}(\mathbf{A} \cdot \mathbf{C})$$

is an **entirely different** vector⁶!

⁶Unlike the scalar triple product $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ where at worst you'll be off by a sign.

The Position Vector



- ▶ The *location* of a point in three dimensions can be described by listing its *Cartesian coordinates* (x, y, z)
- ▶ The *vector*⁷ to that point from the *origin* is called the **position vector**,

$$\mathbf{r} \equiv x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

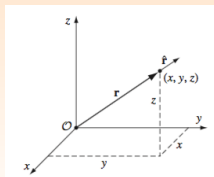
- ▶ Its *magnitude*,

$$r = \sqrt{x^2 + y^2 + z^2}$$

_____ is simply the distance from the origin $\mathcal{O}$.

⁷The *position vector* is not truly a vector in the rigorous sense since we'd mentioned earlier that vectors be freely transposed, while the *position vector* is tethered to the origin. To this extent it's more aptly a **pseudo-vector**.

The Position Vector



- The unit vector

$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{r}$$

points *radially outward*.

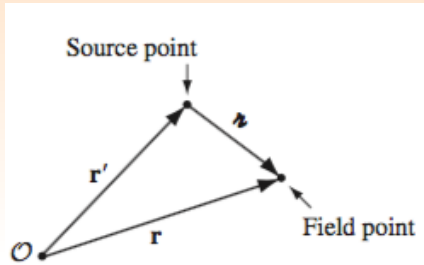
- The **infinitesimal displacement vector**⁸ from (x,y,z) to $(x+dx,y+dy,z+dz)$ is,

$$d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$$

- In order to have $d\mathbf{l}$ point **radially outward** we must have $dx:dy:dz = x:y:z$, i.e.,
 dx, dy, dz must have the same relative proportions as x, y, z .

⁸ $d\mathbf{l}$ does not point in *any particular* direction since $dx \neq dy \neq dz$, in general.

The Separation Vector $\mathbf{s}$



- ▶ In *electrodynamics*, one frequently encounters problems involving two points, typically, a **source point**, $\mathbf{r}'$, where an electric charge is located, and a **field point**, $\mathbf{r}$, at which you are calculating the electric or magnetic field.
- ▶ The **separation vector** $\mathbf{s}$ from the source point to the field point is then,

$$\mathbf{s} \equiv \mathbf{r} - \mathbf{r}' = (x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}$$

Ordinary Derivative

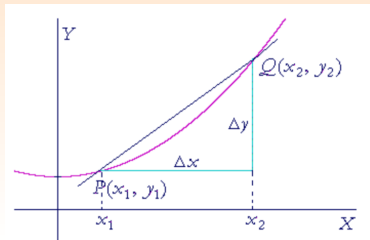


Figure: www.themathpage.com

- ▶ Given a function $f(x)$, the ordinary derivative df/dx represents the rate of change of f w.r.t x .
- ▶ *Alternatively*, it tells us how rapidly f varies when we change x by an infinitesimal (tiny) fraction dx .
- ▶ *Geometrically*, it gives us the **slope** of the graph of f vs. x .

The Partial Derivative

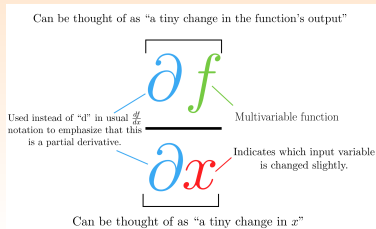


Figure: www.khanacademy.com

- ▶ The **partial derivative** of a function of *several* variables is its derivative with respect to **only one** of those variables, with the others held constant (as opposed to the **total derivative**, in which all variables are allowed to vary).
- ▶ Q: Given $f(x, y) = x^2 y^3$, compute $\frac{\partial f}{\partial y}$, and $\frac{df}{dy}$.
- ▶ Ans: $\frac{\partial f}{\partial y} = 3x^2 y^2$ and $\frac{df}{dy} = 3x^2 y^2 + 2xy^3 \frac{dx}{dy}$

The Del ∇ Operator

- ▶ The **del** operator is *defined* as,

$$\nabla \equiv \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

- ▶ Note, ∇ is not a vector! Instead, it's an operator, which in the *grad(ient)* case, maps a differentiable scalar function f to a vector function ∇f .
- ▶ Nor does it *multiply* what appears on the right of it. (∇T ??)
- ▶ Instead, it's a *vector operator*, or an **instruction** to act on whatever appears to its right.
- ▶ It's really just (very) clever notation⁹ and acts in 3 ways:
 - ▶ On a scalar function, say T : ∇T (*gradient*)
 - ▶ On a vector function, say $\mathbf{v}$, via the dot product: $\nabla \cdot \mathbf{v}$ (*divergence*)
 - ▶ On a vector function, say $\mathbf{v}$, via the cross product: $\nabla \times \mathbf{v}$ (*curl*)

⁹So clever that it allows us to use ∇ "like a vector", but being an operator it alone doesn't have a meaning!

The Gradient

- ▶ Given a **scalar function**, say $T(x, y, z)$, the change in the quantity T when x, y, z are varied **infinitesimally** i.e., by (dx, dy, dz) ¹⁰ is given by the **total differential**,

$$\begin{aligned}dT &= T(x + dx, y + dy, z + dz) - T(x, y, z) \\&= \left(\frac{\partial T}{\partial x}\right)dx + \left(\frac{\partial T}{\partial y}\right)dy + \left(\frac{\partial T}{\partial z}\right)dz \equiv (\nabla T) \cdot (d\mathbf{l})\end{aligned}$$

where

$$\nabla T \equiv \frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}}$$

is, *by definition*, the **gradient** of T , and is evidently a **vector function**.

- ▶ Using then the definition of the **dot product** we can of course also write,

$$(\nabla T) \cdot (d\mathbf{l}) = |\nabla T| |d\mathbf{l}| \cos\theta$$

¹⁰What if the changes in x, y, z were larger (i.e., not infinitesimal)? As a hint consider $f(x)$ and expand it about the point x_0 via a Taylor series as

$$f(x) = f(x_0) + \frac{\partial f}{\partial x} \Delta x + \frac{1}{2!} \frac{\partial^2 f}{\partial x^2} (\Delta x)^2 + \dots$$

The Gradient and the Directional Derivative

- ▶ We may *think* of the gradient as "*a multi-variable generalization of the derivative*".
- ▶ To make this concrete, consider the definition for the **directional derivative** of a function $f(x, y, z, \dots)$ along an *arbitrary* unit vector $\hat{\mathbf{u}}$ as,

$$D_{\hat{\mathbf{u}}}f(\mathbf{a}) \equiv \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\hat{\mathbf{u}}) - f(\mathbf{a})}{h} = \underbrace{\nabla f(\mathbf{a})}_{\text{the gradient of } f \text{ at } \mathbf{a}} \cdot \hat{\mathbf{u}}$$

which is the rate of change of f at the point $\mathbf{a}$ in the direction $\hat{\mathbf{u}}$.

- ▶ Now, ask yourself '*Along which direction is $D_{\hat{\mathbf{u}}}f(\mathbf{a})$ maximised?*'
- ▶ The **directional derivative** can be rewritten as,

$$D_{\hat{\mathbf{u}}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \hat{\mathbf{u}} = |\nabla f(\mathbf{a})| \cos \theta$$

so the largest the $D_{\hat{\mathbf{u}}}f(\mathbf{a})$ can be is when $\theta = 0$, *i.e.*, when $\hat{\mathbf{u}}$ is in the direction of the gradient $\nabla f(\mathbf{a})$.

- ▶ Thus the **gradient** $\nabla f(\mathbf{a})$ **points in the direction of the greatest increase of f , *i.e.*, the direction of steepest ascent.**

Properties of the Gradient

- ▶ The gradient ∇T , or *alternatively*, $\frac{\nabla T}{|\nabla T|}$ at an arbitrary location points in the direction of **maximum increase** of the function T around that particular location/point.
- ▶ The magnitude $|\nabla T|$ gives the slope (rate of increase) along this maximal direction.
- ▶ When the *gradient vanishes*, i.e., $\nabla T = 0$ at (x_i, y_i, z_i) then the *entire* set, $\{(x_i, y_i, z_i)\}$ represent the **stationary** or **critical points** of the function $T(x, y, z)$.
- ▶ These stationary/critical points are either:
 - ▶ **local** maxima (extremum)
 - ▶ **local** minima (extremum)
 - ▶ saddle/minimax points
- ▶ If you want to *locate* the **critical points(s)** of a scalar function of two (or more) variables, simply set its gradient to zero.

Exercise: The Gradient Of a Hill

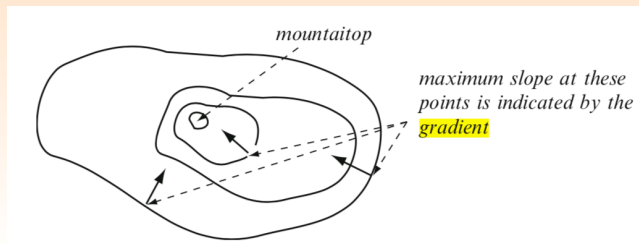
- ▶ **First**, assume a very simple **hill** of elevation,

$$h(x, y) = -x^2 - y^2 + 100$$

having a height 100 m. What do the gradient and the contour lines look like?

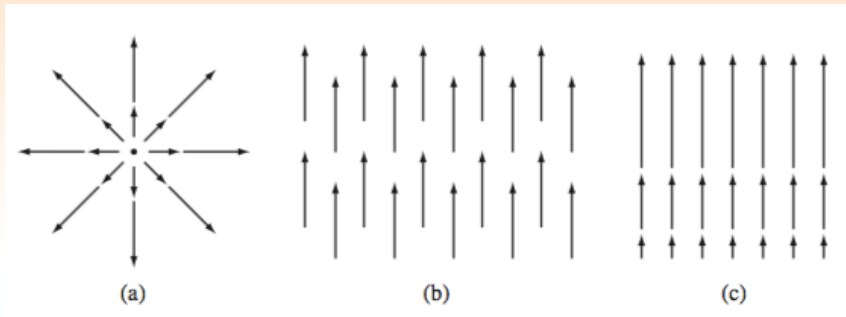
- ▶ **Second**, draw a more arbitrary, realistic hill using **contour lines** (*topological map*).
- ▶ Q:What does the **gradient** $\nabla h(x, y)$ look like?

The Gradient Of a Hill



- ▶ On a *contour/topological map*, the gradient points in a direction which **minimizes** the distance between **adjacent** contour lines. Thus the gradient is always $\perp$ to a contour line.
- ▶ Do all the gradient vectors point exactly toward the **global peak**?
- ▶ **Ans: No, they do not!** The gradient $\nabla h(x, y)$ is a **local quantity** and only gives the magnitude and the direction of maximum increase around the given point (*i.e.*, locally) in question, say (x_i, y_i) .
- ▶ Dwell on the **mnemonic**: "*water always flows in the direction **opposite** the direction of the gradient.*"

The Divergence

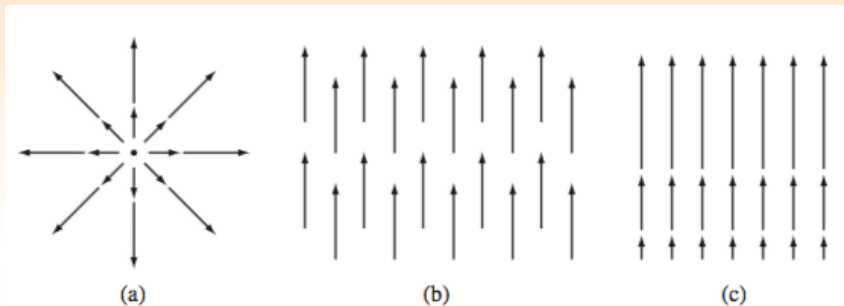


- ▶ The **divergence** of a vector function $\mathbf{v}$ in 3d-space is defined as:

$$\nabla \cdot \mathbf{v} \equiv \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

- ▶ The divergence $\nabla \cdot \mathbf{v}$ is itself a **scalar** function.
- ▶ Is $\nabla \cdot \mathbf{v}$ defined for each point, or do we have one value per $\mathbf{v}$?

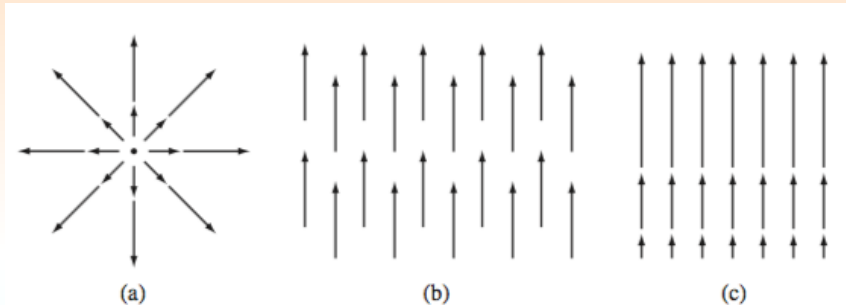
The Divergence



- ▶ The diagrams above plot a **vector field**¹¹ with the (scaled) magnitude and direction of the vector valued function $\mathbf{v}$ at selected points/*grid* in Cartesian space.
- ▶ *Geometrically*, the divergence measures how much the vector $\mathbf{v}$ **spreads out** (diverges) from the point in question.

¹¹In *vector calculus*, a **vector field** is an assignment of a vector to each point in a subset (grid point)) of space.

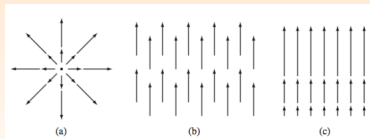
An Intuitive Picture of Divergence



- ▶ Consider a **liquid-flow analogy**. If you drop some **sawdust**¹² at a particular location on the surface and they seem to spread out(in), as opposed to staying stationary, or simply translate, we have location of positive(negative) divergence.
- ▶ A point of positive divergence is a **source**, or faucet; a point of negative divergence is a **sink**, or drain.

¹²Why sawdust?

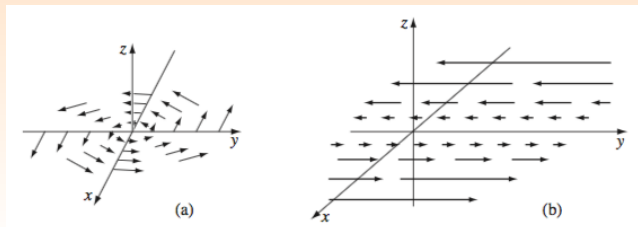
The Divergence: be careful visualizing!



- ▶ One might be lulled into thinking that Fig.(a) represents the $\mathbf{E}$ due to a single, static charge. However, it cannot possibly represent $\mathbf{v} = k \frac{\hat{\mathbf{r}}}{r^2}$, since the radial vectors above are getting longer!
- ▶ Instead, say if Fig.(a) represents $\mathbf{v} = r \hat{\mathbf{r}}$, then it has a constant divergence¹³ everywhere, *i.e.*, $\nabla \cdot \mathbf{v} = 3$.
- ▶ You might think that the vector function given by Fig.(c) has a zero divergence everywhere. But that would be **incorrect**, since the arrows/vectors, starting from the bottom are getting longer.

¹³In spherical coordinates $\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 v_r) + \dots$

The Curl

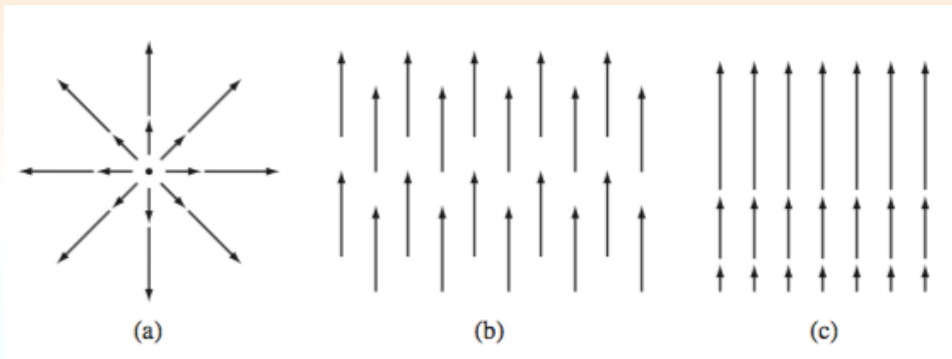


- ▶ The **curl** of a vector function $\mathbf{v}$ is defined as,

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} \quad (\text{vector function})$$

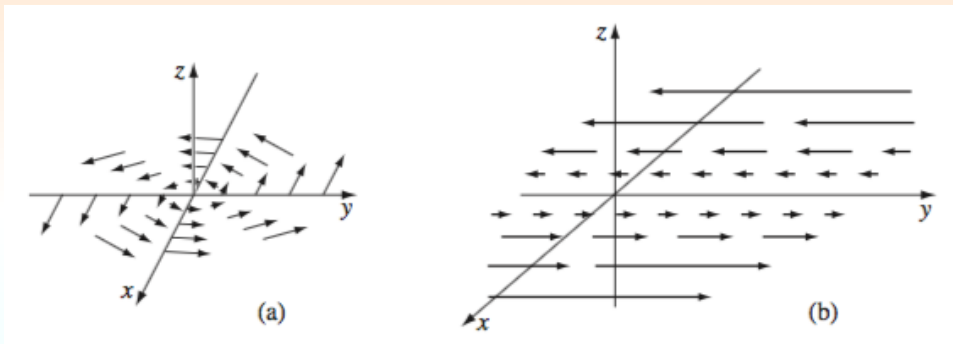
- ▶ *Geometrically*, it measures of how much $\mathbf{v}$ **swirls** around the point in question.
- ▶ *Intuitively*, imagine standing at the edge of a pond. Float a small and light **paddlewheel**; if it starts to **rotate** at that location, then you've placed it at a point of *non-zero* curl.

The Curl



- The three vector fields above all have **zero** curl.

The Curl



- Whereas the functions above have a *substantial* curl, pointing in the $\hat{z}$ direction, as the natural **right-hand rule** would suggest.

Addition and Scalar Multiplication Rules

- ▶ Just like with ordinary calculus, certain mathematical **rules** hold for *vector derivatives*,

$$\nabla(f + g) = \nabla f + \nabla g, \quad \nabla \cdot (\mathbf{A} + \mathbf{B}) = (\nabla \cdot \mathbf{A}) + (\nabla \cdot \mathbf{B}),$$

$$\nabla \times (\mathbf{A} + \mathbf{B}) = (\nabla \times \mathbf{A}) + (\nabla \times \mathbf{B}),$$

and

$$\nabla(kf) = k\nabla f, \quad \nabla \cdot (k\mathbf{A}) = k(\nabla \cdot \mathbf{A}), \quad \nabla \times (k\mathbf{A}) = k(\nabla \times \mathbf{A}),$$

- ▶ The above rules are fortunately quite *intuitive*!

Product Rules

- ▶ While the product rules get a little more *complicated*...¹⁴

Accordingly, there are *six* product rules, two for gradients:

$$(i) \quad \nabla(fg) = f\nabla g + g\nabla f,$$

$$(ii) \quad \nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A},$$

two for divergences:

$$(iii) \quad \nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f),$$

$$(iv) \quad \nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}),$$

and two for curls:

$$(v) \quad \nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f),$$

$$(vi) \quad \nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}).$$

¹⁴Proving these rules once is advised.

Second Derivatives

- By applying ∇ **twice**, we can construct five species of **second derivatives**

(1) Divergence of gradient: $\nabla \cdot (\nabla T)$.

(2) Curl of gradient: $\nabla \times (\nabla T)$.

The divergence $\nabla \cdot \mathbf{v}$ is a *scalar*—all we can do is take its *gradient*:

(3) Gradient of divergence: $\nabla(\nabla \cdot \mathbf{v})$.

The curl $\nabla \times \mathbf{v}$ is a *vector*, so we can take its *divergence* and *curl*:

(4) Divergence of curl: $\nabla \cdot (\nabla \times \mathbf{v})$.

(5) Curl of curl: $\nabla \times (\nabla \times \mathbf{v})$.

(I) The Divergence of a Gradient *aka* The Laplacian

- ▶ When given a scalar function T , the **Laplacian** is defined as,

$$\underbrace{\nabla^2 T}_{\text{just shorthand}} \equiv \nabla \cdot (\nabla T) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}$$

- ▶ Note, the Laplacian of a scalar function T is a **scalar** function!
- ▶ On the other hand, given a vector function $\mathbf{v}$, the **Laplacian** is then *defined* as,

$$\underbrace{\nabla^2 \mathbf{v}}_{\text{just shorthand}} \equiv (\nabla^2 v_x) \hat{\mathbf{x}} + (\nabla^2 v_y) \hat{\mathbf{y}} + (\nabla^2 v_z) \hat{\mathbf{z}}$$
$$\neq \nabla \cdot (\nabla \mathbf{v})$$

and is a **vector** function.

(2) The curl of a gradient

- ... is always zero. *i.e.*,¹⁵

$$\nabla \times (\nabla T) = \mathbf{0}$$

¹⁵Do the proof!

(3) Gradient of a Divergence



$$\nabla(\nabla \cdot \boldsymbol{v})$$

... doesn't show up much in the study of electromagnetism so we won't bother about it.

(4) The divergence of a curl

- ... is also always zero, *i.e.*,¹⁶

$$\nabla \cdot (\nabla \times \boldsymbol{v}) = 0$$

¹⁶Do the proof!

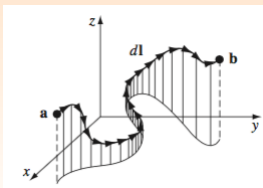
(5) The curl of a curl



$$\nabla \times (\nabla \times \boldsymbol{v}) = \underbrace{\nabla(\nabla \cdot \boldsymbol{v})}_{(3)} - \underbrace{\nabla^2 \boldsymbol{v}}_{(1)}$$

is just the gradient of a divergence minus the Laplacian of a vector, or (3) – (1).

Line Integrals



- ▶ A **line integral** is an expression of the form

$$\int_a^b \underbrace{\mathbf{v} \cdot d\mathbf{l}}_{\text{dot product}}$$

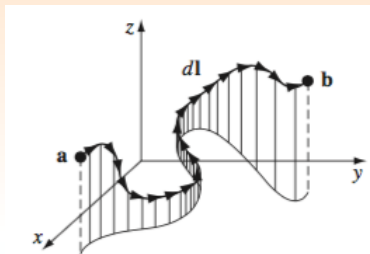
where $\mathbf{v}$ is a vector function, and $d\mathbf{l}$ is the infinitesimal displacement vector.

- ▶ The integral **must be** carried out **along a prescribed path** $\mathcal{P}$ from point $\mathbf{a}$ to point $\mathbf{b}$, and we should instead write

$$\int_a^b \mathcal{P} \mathbf{v} \cdot d\mathbf{l}$$

explicitly mentioning the path $\mathcal{P}$.

Line Integrals

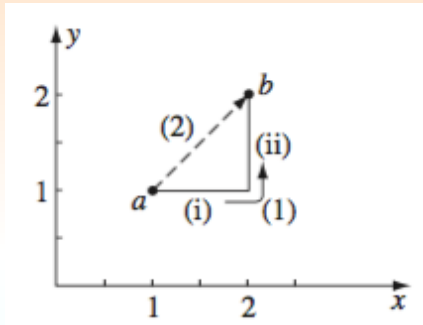


- If the path $\mathcal{P}$ is a **closed loop** (i.e., $\mathbf{a} = \mathbf{b}$) we put a **circle** around the integral sign as,

$$\oint_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l} \neq 0$$

where it's important to note that, in general, closed line integrals don't have to be zero, *in general!*

A Line Integral Example

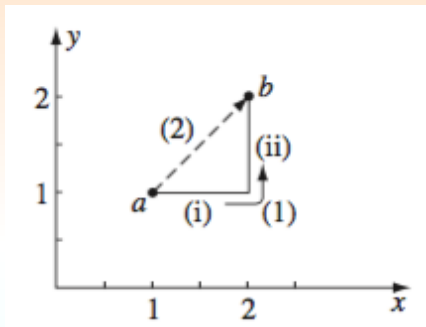


- **Problem:** Calculate the line integral for the vector function

$$\mathbf{v} = y^2 \hat{\mathbf{x}} + 2x(y+1) \hat{\mathbf{y}}$$

from the point $a = (1, 1, 0)$ to $b = (2, 2, 0)$, along the paths (1) and (2) in the above figure.

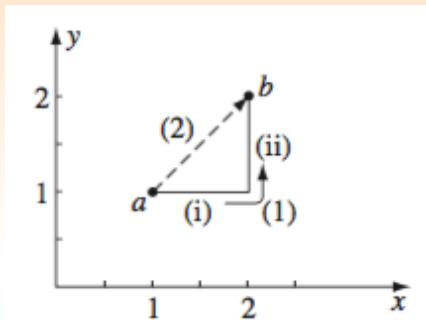
A Line Integral Example: Along path (I)



- Along the **horizontal segment of (I)** (only x varies) $dy = dz = 0$, so

$$d\mathbf{l} = dx\hat{\mathbf{x}}, y = 1, \mathbf{v} \cdot d\mathbf{l} = y^2 dx, \int \mathbf{v} \cdot d\mathbf{l} = (1)^2 \int_1^2 dx = \mathbf{1}$$

A Line Integral Example: Along path (I)



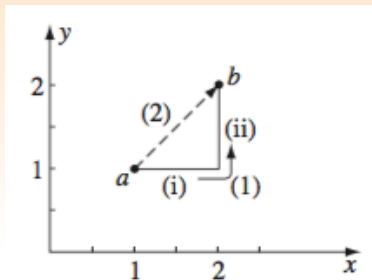
- For the **vertical segment of (I)** (only y varies) $dx = dz = 0$, so

$$d\mathbf{l} = dy\hat{y}, x = 2, \mathbf{v} \cdot d\mathbf{l} = 2x(y+1)dy, \int \mathbf{v} \cdot d\mathbf{l} = 4 \int_1^2 (y+1)dy = 10$$

- Thus, summing the horizontal and vertical parts of (I) we get,

$$\int_{(I)} \mathbf{v} \cdot d\mathbf{l} = 1 + 10 = 11$$

A Line Integral Example: Along path (2)



- For path (2), $x = y \Rightarrow dx = dy$ ¹⁷ and, $dz = 0$

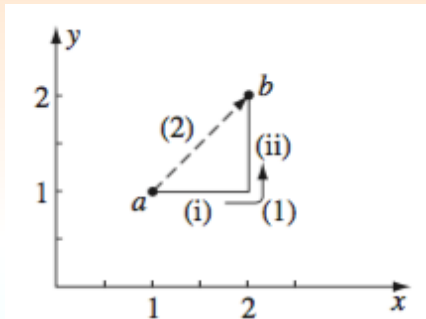
$$d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}}, \mathbf{v} \cdot d\mathbf{l} = x^2 dx + 2x(x+1)dx = (3x^2 + 2x)dx,$$

, and finally,

$$\int_{(2)} \mathbf{v} \cdot d\mathbf{l} = 10$$

¹⁷This step is crucial for a *general* path where variables are interdependent.

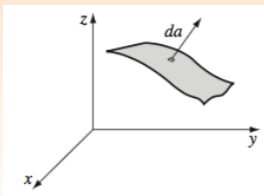
A Line Integral Example: $a \rightarrow (1) \rightarrow (2) \rightarrow a$



- For the *circuitous* path $a \rightarrow (1) \rightarrow (2) \rightarrow a$, we simply **sum** the contributions of paths (1) and (2) and get,

$$\oint \mathbf{v} \cdot d\mathbf{l} = 11 - 10 = \mathbf{1}$$

Surface/Double and Flux Integrals



- ▶ A **surface/double integral** is an expression of the form

$$\int_{\mathcal{S}} f \, da$$

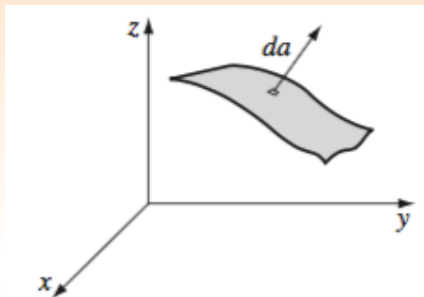
where f is a scalar function, da is an infinitesimal patch of area over a prescribed surface $\mathcal{S}$.

- ▶ However, the **flux**¹⁸ of a vector function $\mathbf{v}$ through $\mathcal{S}$ is defined as,

$$\int_{\mathcal{S}} \mathbf{v} \cdot \underbrace{\mathbf{a}}_{\text{dot product}} \, da$$

¹⁸in analogy with liquid flow, i.e., if $\mathbf{v}$ describes the flow a mass of liquid per unit area per unit time.

Flux Integrals

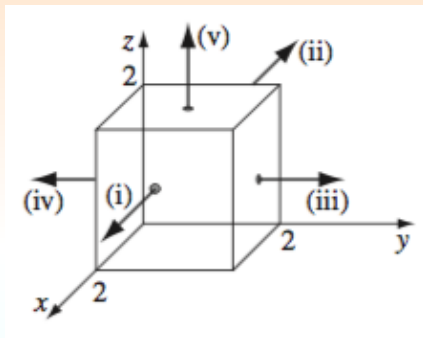


- ▶ Since there are two opposite directions for the surface normal, we *choose* the direction that points **radially outward** with the origin as a reference point.
- ▶ If the surface is **closed**¹⁹ we write,

$$\oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a} \neq 0$$

¹⁹A closed surface in 3-d is exemplified by a *balloon*.

A Flux Example

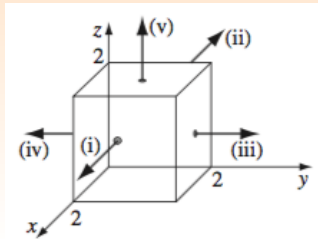


- **Problem:** Calculate the flux

$$\mathbf{v} = 2xz\hat{\mathbf{x}} + (x+2)\hat{\mathbf{y}} + y(z^2 - 3)\hat{\mathbf{z}}$$

over five sides (*excluding the bottom*).

A Flux Example: Surface 1



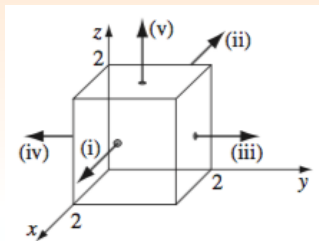
(i) $x = 2$, $d\mathbf{a} = dy dz \hat{\mathbf{x}}$, $\mathbf{v} \cdot d\mathbf{a} = 2xz dy dz = 4z dy dz$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = 4 \int_0^2 dy \int_0^2 z dz = 16.$$

- Note, that for surface (i) variables x and y are independent of each other²⁰, and the integrals for each can be carried out independently.

²⁰Why?

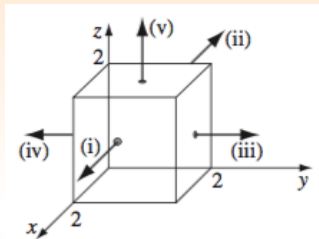
A Flux Example: Surface 2



(ii) $x = 0$, $d\mathbf{a} = -dy\,dz\,\hat{\mathbf{x}}$, $\mathbf{v} \cdot d\mathbf{a} = -2xz\,dy\,dz = 0$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = 0.$$

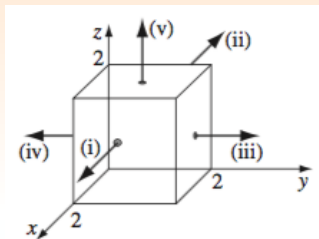
A Flux Example: Surface 3



(iii) $y = 2$, $d\mathbf{a} = dx \, dz \, \hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{a} = (x + 2) \, dx \, dz$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 (x + 2) \, dx \int_0^2 dz = 12.$$

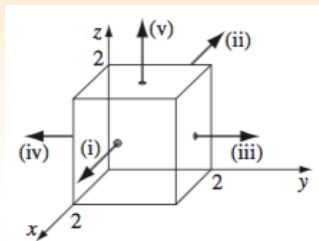
A Flux Example: Surface 4



(iv) $y = 0$, $d\mathbf{a} = -dx\,dz\,\hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{a} = -(x+2)\,dx\,dz$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = -\int_0^2 (x+2)\,dx \int_0^2 dz = -12.$$

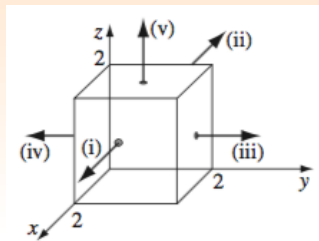
A Flux Example: Surface 5



(v) $z = 2$, $d\mathbf{a} = dx dy \hat{\mathbf{z}}$, $\mathbf{v} \cdot d\mathbf{a} = y(z^2 - 3) dx dy = y dx dy$, so

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 dx \int_0^2 y dy = 4.$$

A Flux Example: Total

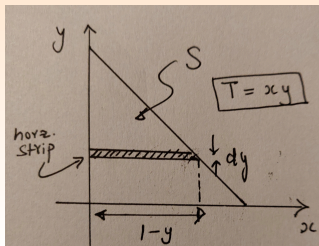


The *total* flux is

$$\int_{\text{surface}} \mathbf{v} \cdot d\mathbf{a} = 16 + 0 + 12 - 12 + 4 = 20.$$

- Note that the surface $\mathcal{S}$ over which the flux is calculated need not be closed.

Evaluating a Surface Integral over a triangular $\mathcal{S}$



- **Problem:** Evaluate

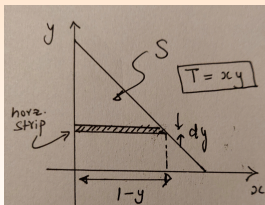
$$\iint_{\mathcal{S}} xy \, dx \, dy$$

over the triangular region with $\perp$ sides of unity, as above.

- **Strategy:**

1. Calculate $T(x, y = \text{const.}) \times$ the area of each infinitesimally short **horizontal strip** of height dy [**INNER SUM**: $x: 0 \rightarrow (1-y)$ ($\iint_{\mathcal{S}} \rightarrow \int_{y=0}^1 \int_{x=0}^{x=1-y}$)], and then,
2. Sum these horizontal strips up one by one [**OUTER SUM**: $y: 0 \rightarrow 1$ ($\iint_{\mathcal{S}} \rightarrow \int_{y=0}^1 \int_{x=0}^{x=1-y}$)].

Evaluating the Surface Integral over a triangular $\mathcal{S}$



- Thus, keeping **y constant**, let's focus now on the [INNER SUM: $x: 0 \rightarrow (1-y)$], i.e.,

$$H(y) = \int_{x=0}^{x=1-y} \underbrace{xy}_{y=\text{const.}} dx = \frac{y(1-y)^2}{2}$$

where the $(1-y)$ limit accounts for the fact that the length of these strips varies.

- Finally, we perform the [OUTER SUM: $y: 0 \rightarrow 1$] and

$$\int_{y=0}^1 H(y) dy = \int_{y=0}^1 \frac{y(1-y)^2}{2} dy = \frac{1}{24}$$

- HW: Find $\int_{\mathcal{S}} xy \, dx dy$ by instead summing **vertical** strips.

Volume Integrals

- ▶ Given a **scalar** function T , volume integrals are expressed as,

$$\int_{\mathcal{V}} T d\tau$$

and $d\tau$ is an infinitesimal volume element.

- ▶ For a **vector** function $\mathbf{v}$, on the other hand, the associated volume integral is,

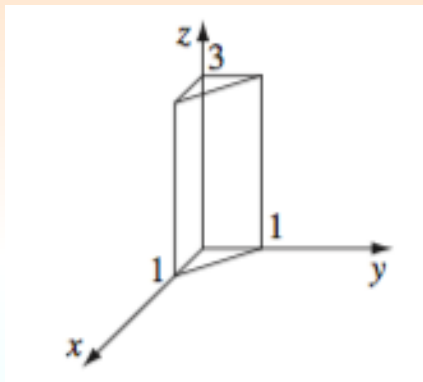
$$\begin{aligned}\int \mathbf{v} d\tau &= \int (v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}) d\tau \\ &= \hat{\mathbf{x}} \int v_x d\tau + \hat{\mathbf{y}} \int v_y d\tau + \hat{\mathbf{z}} \int v_z d\tau\end{aligned}$$

- ▶ In **Cartesian coordinates**,²¹

$$d\tau = dx dy dz$$

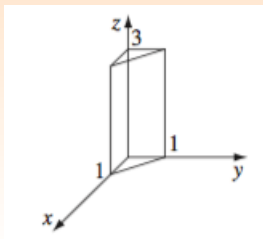
²¹Spherical and cylindrical coordinates are other candidates.

A Volume Integral Example



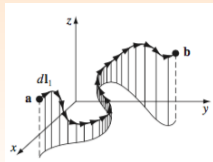
- **Problem:** Calculate the volume integral of $T = xyz^2$.
- It's easy to see from the prismatic volume that z does not depend on either x or y , and ranges from 0 to 3.
- This means that the integral over z can be **factored out**.

A Volume Integral Example



$$\begin{aligned}\int xyz^2 d\tau &= \int_0^3 z^2 dz \underbrace{\iint xy \, dx dy}_{\text{done earlier!}} \\ &= 9 \int_{y=0}^1 y \left[\int_{x=0}^{1-y} x \, dx \right] dy \\ &= \frac{9}{2} \int_0^1 y(1-y)^2 dy = \frac{3}{8}\end{aligned}$$

The Fundamental Theorem for Gradients



- ▶ Given a **scalar** function $T(x, y, z)$, by changing x , y and z *infinitesimally*, i.e., by dx , dy and dz , the variation in T is,

$$dT = (\nabla T) \cdot d\mathbf{l}$$

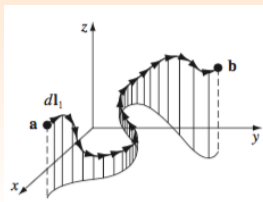
- ▶ If we keep advancing from a to b along units of $d\mathbf{l}_i$ we can **accumulate** the total change in the scalar function T ,

$$\int_a^b (\nabla T) \cdot d\mathbf{l} = T(b) - T(a)$$

also known as the **fundamental theorem for gradients**²².

²²There is also a direct proof using the integral in the LHS above.

The Fundamental Theorem for Gradients



- ▶ Remarkably, the RHS, i.e., $T(b) - T(a)$ makes no reference to the actual **path** taken.
- ▶ It implies that, $\int_a^b (\nabla T) \cdot d\mathbf{l}$ is **independent** of the path taken from a to b .²³
- ▶ It also implies that,

$$\oint (\nabla T) \cdot d\mathbf{l} = 0$$

²³In practice, even though the integral is independent of the path, we **must** pick a specific (if *convenient*) route in order to evaluate it explicitly.

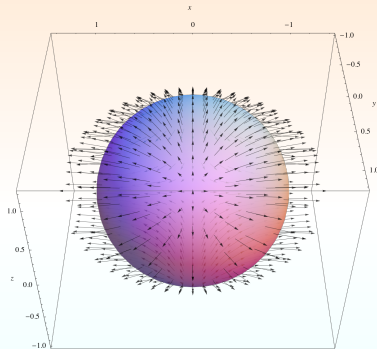
The Fundamental Theorem for Divergences

- ▶ The fundamental theorem for divergences states that:

$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{v}) d\tau = \oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a}$$

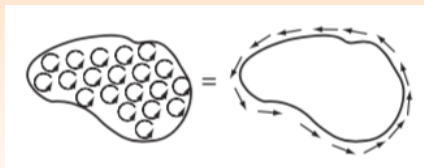
- ▶ It is alternatively known as, **Gauss' theorem**, **Green's theorem**, or simply the **divergence theorem**.
- ▶ **Physical Analogy:** Imagine we're interested in measuring the flux (RHS) of an *incompressible* fluid through a **closed** area $\mathcal{S}$.
- ▶ The divergence theorem states that instead of measuring the flux directly, we could've *equivalently*, sum up all the (liquid) **sources** inside the volume $\mathcal{V}$ enclosed by the surface $\mathcal{S}$.

The Fundamental Theorem for Divergences



- Q: Verify the divergence theorem for $\mathbf{v} = 2x\hat{\mathbf{x}} + y^2\hat{\mathbf{y}} + z^2\hat{\mathbf{z}}$ for the sphere of radius unity. [Ans: LHS = RHS = $\frac{8\pi}{3}$]

The Fundamental Theorem for Curls



- ▶ Also known as **Stoke's theorem** states that,

$$\int_{\mathcal{S}} (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l}$$

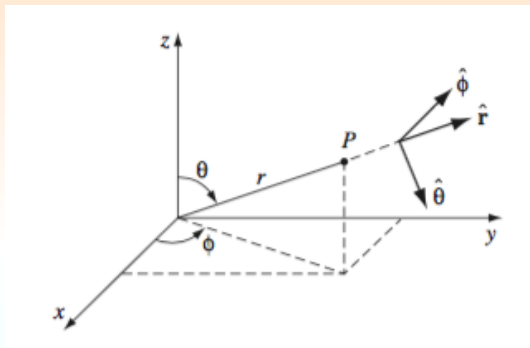
which says that the integral of a curl (of a vector function) over a surface $\mathcal{S}$ is equal to the value of the function integrated over the boundary $\mathcal{P}$) *enclosing* that surface.

- ▶ Sticking with the liquid analogy, since the curl measures the *twist* of $\mathbf{v}$, if we are interested in the **total swirl**, we can equivalently just measure how much the flow $\mathbf{v}$ follows the closed boundary $\mathcal{P}$ enclosing the surface $\mathcal{S}$.
- ▶ As earlier, by convention we select the orientation of $d\mathbf{a}$ pointing outward, and the sense of the line $d\mathbf{l}$ to be anti-clockwise.

The Fundamental Theorem for Curls

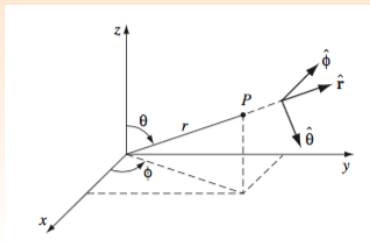
- ▶ As a consequence of the theorem, $\int_{\mathcal{S}} (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ depends only on the boundary line $\mathcal{P}$, **but not on the particular surface $\mathcal{S}$ used**, as long as it's circumscribed by $\mathcal{P}$. Think of an intact **soap bubble** across a fixed loop. It doesn't matter whether the bubble is convex, concave, or combinations thereof, as long as the loop circumscribing it is fixed.
- ▶ As a consequence we may **deform** $\mathcal{S}$ for *mathematical convenience* as long as it satisfies the boundary $\mathcal{P}$.
- ▶ $\oint_{\mathcal{S}} (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$ for **any closed surface**, since the boundary line, like the mouth of a balloon, shrinks down to a point.
- ▶ HW: Do example 1.11 of [IEDJ].

Spherical Coordinates



- An *alternative* to using the usual Cartesian coordinates described by (x, y, z) is to use **spherical coordinates** described by (r, θ, ϕ) where,
 1. r is the distance from the origin (the magnitude of the position vector $\mathbf{r}$)
 2. θ (the angle down from the z axis) is called the **polar angle**
 3. ϕ (the angle around from the x axis) is the **azimuthal angle**.

Spherical Coordinates



- From the above figure we observe that,

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

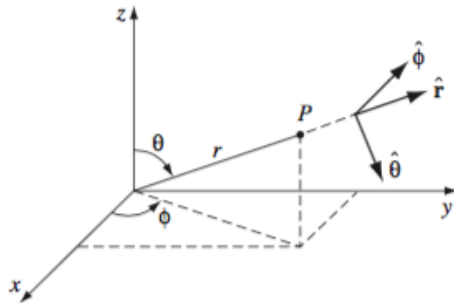
$$z = r \cos \theta$$

- A general vector $\mathbf{A}$ can be represented in spherical coordinates as,

$$\mathbf{A} = A_r \hat{\mathbf{r}} + A_\theta \hat{\boldsymbol{\theta}} + A_\phi \hat{\boldsymbol{\phi}}$$

where $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ form an **orthogonal** basis set.

Spherical Coordinates



- In terms of Cartesian unit vectors we can write,²⁴,

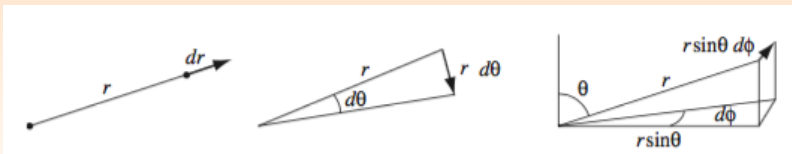
$$\hat{\mathbf{r}} = \sin\theta \cos\phi \hat{\mathbf{x}} + \sin\theta \sin\phi \hat{\mathbf{y}} + \cos\theta \hat{\mathbf{z}}$$

$$\hat{\boldsymbol{\theta}} = \cos\theta \cos\phi \hat{\mathbf{x}} + \cos\theta \sin\phi \hat{\mathbf{y}} - \sin\theta \hat{\mathbf{z}}$$

$$\hat{\boldsymbol{\phi}} = -\sin\phi \hat{\mathbf{x}} + \cos\phi \hat{\mathbf{y}}$$

²⁴Do the proof.

The infinitesimal elements in spherical coordinates



- Beware that in the spherical coordinate system,

$$d\mathbf{l} \neq dr \hat{r} + d\theta \hat{\theta} + d\phi \hat{\phi}$$

$$d\tau \neq dr d\theta d\phi$$

- An infinitesimal displacement in the $\hat{r}$ direction is simply dr , and thus,

$$dl_r = dr$$

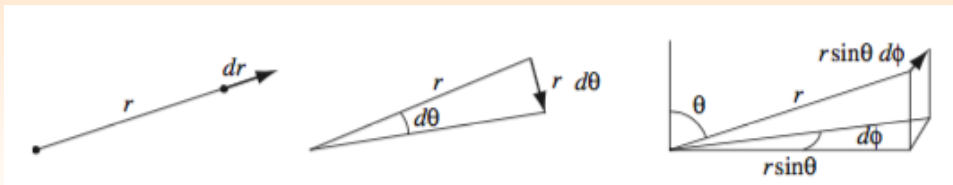
- An infinitesimal element of length in the $\hat{\theta}$ direction is,

$$dl_\theta = r d\theta$$

- An infinitesimal element of length in the $\hat{\phi}$ direction is,

$$dl_\phi = r \sin \theta d\phi$$

The infinitesimal elements in spherical coordinates



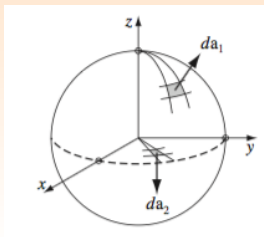
- ▶ Thus the general **infinitesimal displacement** $d\mathbf{l}$ is,

$$d\mathbf{l} = dr\hat{\mathbf{r}} + r d\theta\hat{\boldsymbol{\theta}} + r \sin\theta d\phi\hat{\boldsymbol{\phi}}$$

- ▶ The **infinitesimal volume element** $d\tau$, in spherical coordinates, is the *product* of the three infinitesimal displacements:

$$d\tau = dl_r dl_\theta dl_\phi = r^2 \sin\theta dr d\theta d\phi$$

The surface element in spherical coordinates



- Suppose you're integrating over the *surface* of a sphere of radius r , here,

$$d\mathbf{a}_1 = dl_\theta dl_\phi \hat{\mathbf{r}} = r^2 \sin\theta d\theta d\phi \hat{\mathbf{r}}$$

- On the other hand, if the *surface* lies in the xy plane, as above²⁵

$$d\mathbf{a}_2 = dl_r dl_\phi \hat{\boldsymbol{\theta}} = r dr d\phi \hat{\boldsymbol{\theta}}$$

²⁵What happened to the $\sin\theta$ term in the RHS of $d\mathbf{a}_2$, i.e., $dl_\phi = r \sin\theta d\phi$?

More Formulae for Spherical Coordinates

Gradient:

$$\nabla T = \frac{\partial T}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{\boldsymbol{\phi}}.$$

Divergence:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_\theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}.$$

Curl:

$$\begin{aligned} \nabla \times \mathbf{v} = & \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta v_\phi) - \frac{\partial v_\theta}{\partial \phi} \right] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_\phi) \right] \hat{\boldsymbol{\theta}} \\ & + \frac{1}{r} \left[\frac{\partial}{\partial r} (r v_\theta) - \frac{\partial v_r}{\partial \theta} \right] \hat{\boldsymbol{\phi}}. \end{aligned}$$

Laplacian:

$$\nabla^2 T = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2}.$$

► Things get complicated²⁶...

²⁶The above formulae will be provided, if required, during the exam, and needn't be memorized.

Cylindrical Coordinates

- Do as HW.

Need for the Dirac Delta Function

- ▶ Given a body of mass M located at $\mathbf{r}_0$, we may express it in terms of its mass density $\rho(\mathbf{r})$ as,

$$M = \int_{\mathcal{V}} \rho(\mathbf{r}) d\tau$$

- ▶ But what does the mass density of a **point mass** located at $\mathbf{r}_0$ look like?
- ▶ It can be unequal to zero only at a single point, *i.e.*,

$$\rho(\mathbf{r}) = 0 \quad \forall \mathbf{r} \neq \mathbf{r}_0$$

- ▶ The volume integral, however,

$$\int_{\mathcal{V}} \rho(\mathbf{r}) d\tau = M \quad (\text{finite})$$

is **finite** provided $\mathbf{r}_0$ lies within the volume $\mathcal{V}$.

Need for the Dirac Delta Function

- ▶ It was a long time coming.
- ▶ Since there was no function that encoded such a property, **Paul A. M. Dirac** **invented one**, writing $\rho(\mathbf{r})$ as,

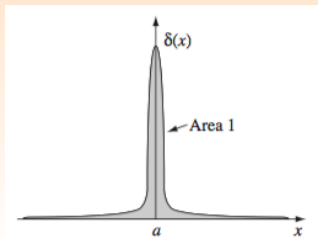
$$\rho(\mathbf{r}) = M \times \delta(\mathbf{r} - \mathbf{r}_0)$$

, and *requiring* that,

$$\int_{\mathcal{V}} \delta(\mathbf{r} - \mathbf{r}_0) d\tau = \begin{cases} 1 & \text{if } \mathbf{r}_0 \in \mathcal{V} \\ 0 & \text{otherwise} \end{cases}$$

$$\delta(\mathbf{r} - \mathbf{r}_0) = 0 \quad \forall \mathbf{r} \neq \mathbf{r}_0$$

The One-Dimensional Dirac Delta Function



- ▶ The one-dimensional **Dirac delta function**²⁷, $\delta(x - a)$, can be pictured as an infinitely high, but infinitesimally narrow spike, with area 1, located at $x = a$, *i.e.*,

$$\delta(x - a) \equiv \begin{cases} 0 & x \neq a \\ \infty & x = a \end{cases}$$

and,

$$\int_{-\infty}^{\infty} \delta(x - a) dx = 1$$

²⁷If x has units of length, what's the unit of $\delta(x)$?

Properties of the One-Dimensional Dirac Delta Function

- ▶ *Technically*, $\delta(x)$ is not a function at all, since its value is not finite at $x=0$.
- ▶ It's **even**, i.e., $\delta(x) = \delta(-x)$.
- ▶ An important characteristic of the Dirac delta function is its **sifting property**²⁸,

$$\int_{-\infty}^{\infty} f(x)\delta(x-x_0)dx = f(x_0) \int_{-\infty}^{\infty} \delta(x-x_0)dx = f(x_0)$$

where, *loosely speaking*, the delta function **picks out** the value of $f(x)$ at $x = x_0$, i.e., $f(x_0)$.

- ▶ Another **curious** property of the Dirac delta function is,

$$\delta(kx) = \frac{1}{|k|}\delta(x)$$

where k is any (non-zero) constant

²⁸Prove the sifting property!

The One-Dimensional Dirac Delta Function

- ▶ If $f(x)$ is differentiable,

$$\int_{-\infty}^{\infty} f(x) \delta'(x - x_0) dx = -f'(x_0)$$

- ▶ The $\delta(x)$ may also be seen as the derivative of the **Heaviside step function**²⁹,

$$\delta(x - a) = \frac{d}{dx} \Theta(x - a)$$

²⁹This property might be invoked while discussing square potential barriers in the QM part of this course.

The Three-Dimensional Dirac Delta Function

- In three dimensions,

$$\underbrace{\delta^3(\mathbf{r} - \mathbf{r}_0)}_{\text{shorthand}} \equiv \delta(x - x_0)\delta(y - y_0)\delta(z - z_0)$$

where $\mathbf{r}_0 = x_0\hat{\mathbf{x}} + y_0\hat{\mathbf{y}} + z_0\hat{\mathbf{z}}$ is the location where $\delta^3(\mathbf{r} - \mathbf{r}_0)$ is centered. and³⁰,

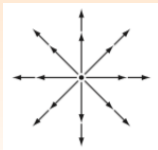
$$\int_{\mathcal{V}} \delta^3(\mathbf{r} - \mathbf{r}_0) d\tau = 1$$

- Also,

$$\int_{\mathcal{V}} f(\mathbf{r}) \delta^3(\mathbf{r} - \mathbf{r}_0) d\tau = f(\mathbf{r}_0)$$

³⁰All you need to ensure is that $\mathbf{r}_0$ lies *within* the volume $\mathcal{V}$.

Divergence of $\hat{\mathbf{r}}/r^2$



- ▶ Consider the vector function³¹,

$$\mathbf{v} = \frac{1}{r^2} \hat{\mathbf{r}}$$

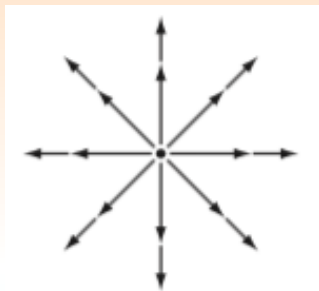
- ▶ From the figure, evidently $\mathbf{v}$ has a **LARGE positive divergence** at the center,
- ▶ ... and **yet...** (do the math)
- ▶

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) = \frac{1}{r^2} \frac{\partial}{\partial r} (1) = 0 \quad \text{everywhere!?!}$$

which is certainly not what we were expecting!

³¹Which is eerily reminiscent of $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}$, the electric field due to a **single, static point** charge.

Divergence of $\hat{\mathbf{r}}/r^2$



- ▶ However, when we consider the **divergence theorem**, *i.e.*,

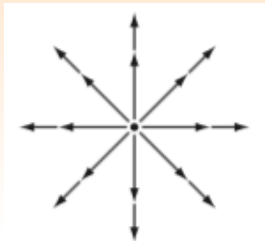
$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a}$$

the RHS is...

- ▶ ...,

$$\oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a} = \int \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \cdot (r^2 \sin\theta d\theta d\phi \hat{\mathbf{r}}) = 4\pi$$

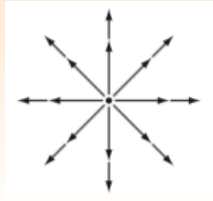
Divergence of $\hat{\mathbf{r}}/r^2$



- ▶ But we'd just (*albeit naïvely*) found that the LHS, i.e., $\int_V (\nabla \cdot \mathbf{v}) d\tau = 0$, which **contradicts** the **divergence theorem**!
- ▶ However, $\int_V (\nabla \cdot \mathbf{v}) d\tau = 0$ is **incorrect** –the **source** of the problem being the point $r = 0$, where $\mathbf{v} = \infty$, i.e., it BLOWS UP.
- ▶ Indeed, $\nabla \cdot \mathbf{v}$ is actually *zero* everywhere **except** the origin.³²

³²**Beware!** The intuition of observing the spreading of sawdust at any of the **non-central** points seems to suggest a non-zero divergence, at least to my eye.

Divergence of $\hat{\mathbf{r}}/r^2$



- ▶ We resolve this *paradox* by realizing that the volume integral of $\nabla \cdot (\hat{\mathbf{r}}/r^2)$ must yield a constant 4π . (Why because the RHS of the **divergence theorem** just said so.)
- ▶ Thus, we can write using [the definition of the Dirac delta function](#),

$$\nabla \cdot (\hat{\mathbf{r}}/r^2) = 4\pi\delta^3(\mathbf{r})$$

- ▶ Alternatively, $\frac{1}{4\pi}\nabla \cdot (\hat{\mathbf{r}}/r^2)$ is a **concrete representation** of the Dirac delta function $\delta^3(\mathbf{r})$.

Proof of the Sifting Property

- Q: Prove the **sifting property** of the Dirac delta function, *i.e.*,

$$\int_{-\infty}^{\infty} f(x)\delta(x)dx = f(0)$$

- We need to consider a **concrete representation** of the Dirac delta function such as the **top-hat function** $T(x)$ defined by

$$T(x) \equiv \begin{cases} \frac{1}{2\epsilon} & |x| \leq \epsilon \\ 0 & |x| > \epsilon \end{cases}$$

which in the $\lim_{\epsilon \rightarrow 0}$ goes into the Dirac delta function. Note that the area under curve of the top-hat function is 1.

- In other words,

$$\delta(x) = \lim_{\epsilon \rightarrow 0} T(x)$$

Proof of the Sifting Property

- Consider

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) T(x) dx &= \frac{1}{2\epsilon} \int_{-\epsilon}^{\epsilon} f(x) dx \\ &= \frac{F(\epsilon) - F(-\epsilon)}{2\epsilon}\end{aligned}$$

where $F(x)$ is the *anti-derivative* of $f(x)$.

- Taking the $\lim_{\epsilon \rightarrow 0}$ we get

$$\lim_{\epsilon \rightarrow 0} \frac{F(\epsilon) - F(-\epsilon)}{2\epsilon} = f(0)$$

A Nice Proof of the Sifting Property

- Recognize that $\delta(x)$ is only *appreciable* over a **tiny region** $x \leq |\epsilon|$, and thus

$$\int_{-\infty}^{\infty} f(x)\delta(x)dx = \int_{-\epsilon}^{\epsilon} f(x)\delta(x)dx$$

- Since the interval $x \leq |\epsilon|$ is tiny, and the function $f(x)$ is well-behaved, we might as well take it nearly constant in the extremely narrow interval around $x = 0$, i.e., $f(0)$. Thus, we may take it out of the integral and use the defining property of $\delta(x)$ that $\int_{-\infty}^{\infty} \delta(x)dx = 1$

$$\int_{-\epsilon}^{\epsilon} f(x)\delta(x)dx = f(0) \int_{-\epsilon}^{\epsilon} \delta(x)dx = f(0)$$

Alternative Representations of $\delta(x)$

optional

- ▶ There are *several* alternative representations of the Dirac delta function.
- ▶ Think about whether $\frac{\sin(x)}{x}$ is a candidate.
- ▶ However, note that $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$ and thus falls short of at least one of the requirements of the Dirac delta function $\delta(x)$, i.e., its value must be ∞ at $x = 0$.
- ▶ However, your guess wasn't far away since $\lim_{\epsilon \rightarrow 0} \frac{1}{\pi} \frac{\sin(\frac{x}{\epsilon})}{x}$ is a valid representation of $\delta(x)$.
- ▶ Another commonly suspected candidate is $\frac{1}{x^2}$.
- ▶ Now while it does blow up at $x = 0$, and is clearly even, it falls short of $\delta(x)$ since isn't zero at all points apart from $x = 0$.
- ▶ Again you were close, and instead the function $\lim_{\epsilon \rightarrow 0} \frac{\epsilon}{x^2 + \epsilon^2}$ is a valid representation of $\delta(x)$

The Scalar Potential V

- ▶ When $\nabla \times \mathbf{E} = 0$ everywhere³³, **Stokes' theorem** tells us that over any & every possible closed path,

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$

- ▶ On the other hand, the **converse** of the **fundamental theorem of gradients**³⁴ allows us to write,

$$\oint -\nabla V \cdot d\mathbf{l} = 0$$

- ▶ Thus, a *static* $\mathbf{E}$ can be written as³⁵ the **gradient of a scalar potential V** ,

$$\mathbf{E} = -\nabla(V + \text{const.})$$

- ▶ The potential is **not unique**, *i.e.*, any **constant** can be added to V **without affecting** its gradient, *i.e.*, the (negative) electric field $-\mathbf{E}$.

³³ $\nabla \times \mathbf{E} = 0$ is guaranteed only in the *electrostatic* regime.

³⁴**Converse** of the fundamental theorem of gradients: "*If the integral of $\mathbf{v}$ over every closed loop in the domain of $\mathbf{v}$ is zero, then $\mathbf{v}$ is the gradient of some scalar-valued function, say V .*"

³⁵The negative sign is purely a matter of convention.

Curl-less (or irrotational) fields $\mathbf{v}_{irr}$

- ▶ The condition $\nabla \times \mathbf{E} = 0$ everywhere is **equivalent** to,
 - ▶ $\int_a^b \mathbf{E} \cdot d\mathbf{l}$ is independent of path, for any given end points, a consequence of the fundamental theorem of gradients.
 - ▶ $\oint \mathbf{E} \cdot d\mathbf{l} = 0$.
 - ▶ $\mathbf{E}$ is the gradient of some scalar function: $\mathbf{E} = -\nabla V$
- ▶ Irrotational fields $\mathbf{v}_{irr}$, i.e., $\nabla \times \mathbf{v}_{irr} = 0$, are also known as **conservative fields**.³⁶
- ▶ We'll soon **prove** in the chapter on **Electrostatics** that the **work done** in moving a test charge against a background of **static** charges is **independent** of the path taken.

³⁶In **mechanics**, a **conservative force** $\mathbf{F} = -\nabla U$ is a force with the property that the total work done $W = \int_a^b \mathbf{F} \cdot d\mathbf{l}$ in moving a particle between point a to b is *independent* of the path taken.

The Vector Potential $\mathbf{A}$

- ▶ Maxwell's equations *guarantee* that

$$\nabla \cdot \mathbf{B} = 0$$

- ▶ This allows $\mathbf{B}$ to be written as the curl of a **vector potential** $\mathbf{A}$,

$$\mathbf{B} = \nabla \times (\mathbf{A} + \nabla T)$$

since mathematically (4) the divergence of a curl is always zero, i.e., $\nabla \cdot (\nabla \times \mathbf{v}) = 0$

- ▶ The vector potential $\mathbf{A}$ is **not unique**, i.e., the gradient of any scalar function $+\nabla T$ can be added to $\mathbf{A}$ without affecting its curl, i.e., the magnetic field $\mathbf{B}$, since mathematically (2) the curl of a gradient is *always zero*, i.e., $\nabla \times (\nabla T) = 0$.

Divergence-less (or solenoidal) fields

- ▶ The condition $\nabla \cdot \mathbf{B} = 0$ everywhere is *equivalent* to,
 - ▶ $\oint \mathbf{B} \cdot d\mathbf{a} = 0$ for any closed surface, a direct consequence of the divergence theorem.
 - ▶ $\mathbf{B}$ is the curl of some vector function: $\mathbf{B} = \nabla \times \mathbf{A}$
 - ▶ $\int_{\mathcal{S}} \mathbf{B} \cdot d\mathbf{a}$ is independent of the details of the precise **open** surface $\mathcal{S}(\mathcal{P})$, once its periphery $\mathcal{P}$ is set.³⁷

³⁷This means we can deform the surface as we please as long as the boundary is kept fixed.

The Helmholtz Theorem

- ▶ "A well-behaved (*i.e.*, goes to zero at infinity) vector field, say $\mathbf{F}$, is **uniquely** specified by its **divergence and curl** (and, in the case of a *finite region*, additionally by its normal component over the entire boundary.)"
- ▶ *i.e.*, suppose we know over all space.³⁸

$$\nabla \cdot \mathbf{F}(\mathbf{r}) = D(\mathbf{r}) \quad \text{and,} \quad \nabla \times \mathbf{F}(\mathbf{r}) = \mathbf{C}(\mathbf{r})$$

- ▶ Then the **unique** vector field $\mathbf{F}$ is given by

$$\mathbf{F}(\mathbf{r}) = -\nabla U(\mathbf{r}) + \nabla \times \mathbf{W}(\mathbf{r}) \quad \text{Helmholtz Decomposition}$$

where,

$$U(\mathbf{r}) = \frac{1}{4\pi} \int_{\text{all space}} d^3 r' \overbrace{\frac{\nabla' \cdot \mathbf{F}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}}^{D(\mathbf{r}')} \quad \text{and,} \quad \mathbf{W}(\mathbf{r}) = \frac{1}{4\pi} \int_{\text{all space}} d^3 r' \overbrace{\frac{\nabla' \times \mathbf{F}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}}^{\mathbf{C}(\mathbf{r}'')}$$

where the ∇' denotes that the derivatives are to be taken *w.r.t.* source points $\mathbf{r}'$.

³⁸Note that since (4): $\nabla \cdot (\nabla \times \mathbf{v}) = 0$, we must have $\nabla \cdot \mathbf{C} = 0$ for consistency.

Why's Helmholtz Theorem Useful?

- ▶ Since a vector field is completely specified once its divergence and curl are known (a purely mathematical result), and we know that the study of electromagnetism involves the vector fields $\mathbf{E}$ and $\mathbf{B}$, we can already *guess*...
- ▶ ... the laws of electromagnetism:

$$\nabla \cdot \mathbf{E} = \text{something}_1$$

$$\nabla \times \mathbf{E} = \text{something}_2$$

$$\nabla \cdot \mathbf{B} = \text{something}_3$$

$$\nabla \times \mathbf{B} = \text{something}_4$$

which look exactly like **Maxwell's equations**³⁹, which are *thus mathematically sufficient* to reconstruct $\mathbf{E}$, and $\mathbf{B}$.

- ▶ Even further, we can *almost solve* for the fields $\mathbf{E}$ and $\mathbf{B}$ via the **Helmholtz decomposition**, even without explicitly knowing what the $RHS = \text{something}_i$ are!

³⁹If you're wondering where $\mathbf{D}$ and $\mathbf{H}$ went, here we're discussing the so-called **microscopic** representation of Maxwell's equations. $\mathbf{D}$ and $\mathbf{H}$ appear in the **macroscopic** formulation where the material medium is built into the equations. Both formulations are *equally* general.

Why's Helmholtz Theorem Useful?

- ▶ Considering the **static**⁴⁰ version of Maxwell's equations

$$\nabla \cdot \mathbf{E} = \frac{\rho(\mathbf{r})}{\epsilon_0}$$

$$\nabla \times \mathbf{E} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j}(\mathbf{r})$$

we're **gratified** by noting that a given set of **stationary** charges $\rho(\mathbf{r})$, and **steady** currents $\mathbf{j}(\mathbf{r})$, may only generate **one** possible steady $\mathbf{E}$, and **one** possible steady $\mathbf{B}$ field.

- ▶ Similarly, if all the sources ρ , and currents $\mathbf{j}$ are **zero everywhere**, then the **only** physical solution is $\mathbf{E} = \mathbf{B} = 0$.
- ▶ This implies that **static electric and magnetic fields cannot generate themselves**, *i.e.*, there must be stationary charges and steady currents generating them!

⁴⁰*i.e.*, all charges are stationary, and currents steady.

Limitations of our Treatment of the Helmholtz Theorem

- ▶ Our discussion for the Helmholtz theorem only works for **time-independent** sources, and currents.
- ▶ For the **time-dependent** case, while the divergence and curl still uniquely identify the vector field, the **Helmholtz decomposition** looks a bit different, though I will not be writing it down explicitly.
- ▶ A second issue is that we haven't grappled with the application of the **boundary conditions** (*i.e., the normal component of the field must be known at all points of the periphery*) that need to be imposed if we're considering a **finite region of space**.